Latent Render-and-Compare: SE(3) Diffusion with a Frozen Feature-Space World Model

Supplementary Material

This supplement gives implementation details (Sec. A), the world-model training procedure (Sec. B) and geometry tokens (Sec. C), the evaluation protocols (Sec. D), our reproduction of LiePose and the choice between averaged and live weights (Sec. E), an audit of the information used at training and test time (Sec. F), and additional results (Sec. G). Section, table and equation numbers without a letter refer to the main paper.

A. Implementation details

Observation encoding. Crops of 224×224 pixels are produced by LiePose's data pipeline, including its training-time augmentation (dynamic zoom-in, colour, blur and noise), and are exported once; DINOv2 ViT-S/14 (frozen, ImageNet normalisation) encodes them into 16×16 patch tokens that are average-pooled 2×2 to 64 tokens of 384 dimensions and stored in half precision. Foreground weights are the visible mask, max-pooled to the token grid and dilated by one token. On SPEED+, two augmented copies of every synthetic training image are exported; on T-LESS, 180,224 augmented training crops are drawn from LiePose's training set (PBR and real images). **[TODO: YCB-V numbers of exported crops.]**

World model. Table 1 lists the hyperparameters. The candidate set per training crop, the reference noise and the loss are described in Sec. B.

Denoiser. Token transformer of width 256, depth 4, 8 heads, MLP ratio 4, adaLN modulation with zero-initialised gates in every block. Input tokens: LayerNorm then a linear map of the concatenation $[\mathbf{z}_p, \hat{\mathbf{z}}_p, \hat{\mathbf{z}}_p - \mathbf{z}_p, w_p]$ ($3 \cdot 384 + 1$ dimensions), plus a learned position embedding. The conditioning token is the sum of Fourier embeddings of the step and of the 6-dimensional Lie-algebra coordinates of the noisy pose and a linear embedding of the crop features (log zoom, normalised crop centre, and on SPEED+ the log box aspect), followed by SiLU and a linear layer. The output head maps the final conditioning token to the 6-dimensional noise.

Diffusion, optimisation, sampling. The generative process is LiePose's SE(3) diffusion formulation, run with its public code and unchanged; it is not a contribution of this paper, and it is shared by LaRC, the control and the LiePose baseline, so that they differ only in the denoiser. As LiePose: SE(3) parametrisation with tangent-space representation, 100 noise levels with a variance-exploding power-law schedule, noise-prediction loss, Adam with learning rate 10^{-4} and exponential decay to 10^{-5} over the second half of training, weight averaging with decay 0.999 every 5 steps, and the 100-step sampler with the same options as LiePose's test script. The world-model parameters are restored after every optimiser step, so that weight decay cannot move them.

B. World-model training

Candidates. For each training crop with pose $g^\star = (R^\star, t^\star)$, a batch row contains: $g^\star$; 11 rotations drawn uniformly on SO(3) with $t^\star$; 12 rotations obtained by rotating $R^\star$ by a random axis and an angle up to $30°$, with $t^\star$; 4 copies of $R^\star$ with $t^\star$ perturbed by Gaussian noise (scaled per axis to the range of the translation parametrisation); 4 copies of $R^\star$ with translations of other training crops. The rotation distance of each candidate to the ground truth is the minimum over the object's symmetric equivalents.

Reference. One reference per row, retrieved at a noisy copy of the ground truth and never from the row's own source image (exclusion is by source image, so augmented copies of the same image are excluded together). All candidates of the row share the reference, as the samples of one image share retrievals at nearby poses during diffusion.

Loss. Equation (3): cross-entropy between the softmax of negative energies over the candidates and the soft target built from the candidate distances, plus the energy at the ground truth. The temperature is a learned scalar initialised at e^{-2}.

SYMSOL-T extras. On SYMSOL-T the world model is trained from scratch on crops, and optionally with two sets of extra regression targets under the weighted-energy loss: in-plane rotations of training images (exact poses obtained by rotating the image), and pairs rendered with the SYMSOL-T renderer at small pose offsets from training poses. The rendered pairs are additional synthetic data that LiePose does not use (Sec. F). Four variants are trained for 30k steps (retrieval only; + geometry tokens; + rolls and pairs; + both), and one is selected on held-out training images (Sec. F); Tab. 4 reports all four. **[TODO: name the selected variant.]**

Table 1. Hyperparameters. WM = world model. **[TODO: complete the YCB-V column.]**

	T-LESS	YCB-V	SPEED+	SYMSOL-T
WM width / depth / heads	384/6/6		384/6/6	384/6/6
WM geometry tokens	no		yes (geo) / no (ret)	yes
WM steps / batch	30k/32		30k/32	20k/32
WM optimiser	AdamW, lr $2 \cdot 10^{-4}$, wd 0.05, cosine, clip 1.0			
candidates (GT/uni/near/t/t-other)	1/11/12/4/4		1/11/12/4/4	1/11/12/4/4
near-rotation range	30°		30°	30°
σ_R / σ_t	10°/0.1		10°/0.1	15°/0.1
reference noise (rot.)	5°		5°	3°
retrieval λ	0		0	0.05
symmetries in bank	BOP models_info		none	analytic
WM checkpoint selection	held-out train images		synthetic val split	final step
denoiser width / depth / heads	256/4/8	256/4/8	256/4/8	256/4/8
steps × batch	400k×64	400k×64	400k×64	400k×32
noisy poses per image	64	64	64	64
LiePose (ours): steps × batch	400k×64	400k×64	400k×64	800k×16
LiePose (ours): noisy poses	256	256	256	256

Checkpoint selection. T-LESS: 3% of the training source images are held out of the bank and of training; the checkpoint with the lowest selection error on 500 of them is kept, and test data are evaluated once. SPEED+: the same criterion on the synthetic validation split. SYMSOL-T: the final step of each variant, scored on held-out training images (Sec. F).

C. Geometry tokens

Let K be the camera intrinsics and (s, x_0, y_0) the zoom and origin of a crop, so that a crop pixel is $s(u-(x_0, y_0))$. A pose (R, t) in LiePose's parametrisation is mapped to the camera frame with the crop box (as in LiePose's translation recovery), including the conversion from allocentric to egocentric rotation. For the reference token p with crop-pixel centre c_p, we back-project its image position onto the plane $Z = Z^{\text{ref}}$ at the reference object depth, P_p, express it in the object frame, $o_p = (R^{\text{ref}})^\top (P_p - t^{\text{ref}})$, move it to the target pose, $P'_p = Ro_p + t$, and project it into the observation's crop. The token receives $[\, c_p/112-1,\ (\pi_k(P'_p)-c_p)/\Delta,\ \log(Z'_p/Z^{\text{ref}})\,]$, with Δ the token size in pixels, Fourier-encoded at frequencies $2^{-2}, \ldots, 2^6$ and added through a zero-initialised MLP. The plane approximation is exact for the object centre and first-order for nearby surface points; it is only an input, never a metric. The seven virtual points are the object centre and the six points at $\pm r$ along the object axes, with r a nominal radius (half the shape size on SYMSOL-T; the largest keypoint radius on SPEED+); their projections at both poses, their image displacement and both depths form seven tokens with learned identity embeddings. Lens distortion is ignored.

D. Evaluation protocols

BOP (T-LESS, YCB-V). We use LiePose's test dataset on the BOP19 test targets and its evaluator, which writes the BOP CSV and calls the BOP toolkit. The toolkit's C++ renderer was not available in our environment, so VSD is computed with the toolkit's Python (vispy) renderer; this changes only how the depth maps used by VSD are rendered. MSSD and MSPD do not use a renderer. **[TODO: state the number of test instances per dataset.]**

SPEED+. Estimates are converted from LiePose's crop-relative parametrisation to the camera frame with the estimated translation and the crop box. For each image, $e_q = 2 \arccos |\langle q, \hat{q} \rangle|$ and $e_t = \|t - \hat{t}\|$. The score is $S_q + S_t$ with $S_q = e_q^{\text{rad}}$ set to zero when $e_q < 0.169°$ and $S_t = e_t/\|t\|$ set to zero when $e_t/\|t\| < 2.173$ mm/m, using the SPEC2021 thresholds [2]; each component is thresholded on its own, and the reported score, rotation part and translation part are means over all images of the lightbox and sunlamp sets. The SPEED+ paper [2] states the thresholds as zeroing the pose score only when both components are below them. **[TODO: check which rule the official SPEC2021 scoring code applies and use it for all rows; the two rules differ only for images with one component below its threshold.]**

SYMSOL-T. For each shape we report the mean over test images of the minimum rotation angle to the set of symmetric ground-truth rotations and of the Euclidean translation error, as LiePose. **[TODO: state the number of test images per shape used.]**

E. Reproducing LiePose and choosing weights

We retrained LiePose with its public code on every benchmark. On T-LESS our run uses batch 64 instead of the paper's 32 (400k steps in both cases), and our models are matched to our run. LiePose's test script samples with the exponential moving average (EMA) of the weights. For our T-LESS checkpoint, the EMA weights differ from the live weights by a small relative amount but give a much higher training loss than the live weights, and sampling with them fails; sampling with the live weights reproduces the published accuracy. We therefore choose, for every checkpoint we report (LiePose, control and LaRC, on every benchmark), between EMA and live weights by the training loss on fresh training batches, before any test evaluation. Table 2 lists the losses and the choices.

Table 2. Training loss of EMA and live weights on fresh training batches, and the weights used for testing. **[TODO: fill; values measured so far are recorded in the source comments. The control and LaRC numbers in the tables so far sample with EMA weights; the training-loss check for them has not been run yet.]**

Benchmark	Model	EMA	live	used
T-LESS	LiePose (ours)	5.39	2.50	live
T-LESS	Control			
T-LESS	LaRC			
YCB-V	LiePose (ours)			
YCB-V	Control			
YCB-V	LaRC			
SPEED+	LiePose (ours)	0.593	0.601	EMA
SPEED+	Control			
SPEED+	LaRC			
SYMSOL-T	LiePose (ours)			
SYMSOL-T	Control			
SYMSOL-T	LaRC			

F. Information audit

At test time. (i) Crops from ground-truth boxes on all benchmarks (Sec. 4.6): visible box and visible mask on T-LESS and YCB-V, as in LiePose; the box of the keypoints projected with the ground-truth pose on SPEED+; the box of the rendered foreground on SYMSOL-T, where LiePose uses the full image. (ii) The object identity, used to choose the reference bank. (iii) Camera intrinsics, for geometry tokens. (iv) Tokens, poses and boxes of training crops, in the bank. No test image is in the bank, and no ground-truth pose is used by the model; ground truth is read only by the scorers and by rows and figures marked as diagnostics (oracle selection, oracle tokens, flip timing, world-model discrimination).

During training. Only training images: PBR and real training images on T-LESS and YCB-V, synthetic images on SPEED+ (the unlabeled lightbox and sunlamp images are not used in any form), and LiePose's training split on SYMSOL-T, plus the in-plane rotations of these images and the rendered pose pairs used by the SYMSOL-T world model (Sec. B).

Model selection. Denoisers: final checkpoint, EMA or live weights by training loss (Sec. E). World models: held-out training images on T-LESS, the synthetic validation split on SPEED+, and on SYMSOL-T 3% of the training images (1,350 per shape, none of them a source of the rolls or rendered pairs), which are removed from the world model's training set and reference bank. The selection score is the refinement pick error on 500 of these images (the energy's choice among 24 rotation perturbations of 0.25–8° around the ground truth); the variant with the lowest score at its final step is used, a rule fixed before training. Test images were not used to train or select any of the four variants. However, the variants themselves were designed during an earlier development phase in which the same diagnostic was computed on 500 test images; we disclose this, and Tab. 4 reports every variant on the test set.

Deviations from LiePose's setup. 64 instead of 256 noisy poses per training image for the control and LaRC; batch 64 for our T-LESS reproduction of LiePose (paper: 32); weights chosen by training loss (Sec. E); VSD rendered with the Python renderer. The control and LaRC train on a fixed set of pre-augmented crops whose tokens are cached (180,224 crops on T-LESS, two augmented copies of each synthetic image on SPEED+), whereas LiePose draws a new augmentation every time an image is used.

G. Additional results

Table 3. T-LESS rotation and translation accuracy (%): share of estimates with symmetric rotation error below 2/5/10 degrees and translation error below 2/5/10 cm, as reported by LiePose. **[TODO: fill.]**

Method	R@2	R@5	R@10	T@2	T@5	T@10
GDRNPP (detected boxes), as reported in [1]	21.6	71.2	90.6	90.3	96.1	98.1
LiePose [1]	47.2	86.9	94.8	71.7	92.0	97.2
LiePose (our training)	50.5	86.7	92.3	74.1	91.7	96.8
Control	11.3	47.1	76.4	34.8	69.5	88.2
LaRC	14.7	54.0	80.4	44.4	79.2	93.8
LaRC + selection						

Per-object results. **[TODO: per-object BOP recalls for T-LESS (30 objects) and YCB-V (21 objects), for LiePose (ours), the control and LaRC, as LiePose's Tab. 8.]**

Table 4. SPEED+ breakdown: rotation and translation parts of the score, median errors, and the share of flipped estimates ($e_q > 90°$). **[TODO: fill.]**

Split	Method	S_q	S_t	med. e_q	med. e_t	flips
lightbox	LiePose (ours)	0.203	0.031	3.0	0.095	4.2%
	Control					
	LaRC (geo.)	0.647	0.055	8.1	0.196	18.2%
	LaRC + sel.					
sunlamp	LiePose (ours)	0.260	0.029	5.0	0.107	5.5%
	Control					
	LaRC (geo.)	0.669	0.047	9.0	0.189	19.7%
	LaRC + sel.					

Table 5. Training budget. The same denoiser at a short and at the matched budget (T-LESS AR %, SPEED+ lightbox S, SYMSOL-T mean rotation error). **[TODO: fill.]**

Model	steps × batch	noisy poses	T-LESS	SPEED+	SYMSOL-T
Control	100k × 32	16	41.2	0.782	
LaRC	100k × 32	16	54.7	0.734	
Control	400k × 64 (32)	64	47.6		
LaRC	400k × 64 (32)	64	54.3		

Table 6. World-model design (discrimination as in Tab. 7, rotation perturbations 1–3° / 3–10°; selection error = rotation error of the lowest-energy candidate, deg). **[TODO: fill.]**

Variant (T-LESS)	1–3°	3–10°	selection error
Per-object learned template (no reference)			
Retrieved reference, reconstruction loss only			
Retrieved reference, $\mathcal{L}_{\text{WM}}$ (LaRC)			
without symmetric equivalents in the bank			
bank restricted to other objects (held-out objects)			

Compute. **[TODO: GPU type and hours per run (world model, denoiser, LiePose), on the local workstation and the cluster.]**

Broader impact. Pose estimation of known objects supports robotic manipulation and autonomous spacecraft rendezvous. We see no specific risk beyond those of perception systems in general; for safety-critical use, the failure modes in Fig. 4 and the reliance on ground-truth crops (Sec. 6) must be addressed first.

References

[1] Tsu-Ching Hsiao, Hao-Wei Chen, Hsuan-Kung Yang, and Chun-Yi Lee. Confronting ambiguity in 6D object pose estimation via score-based diffusion on SE(3). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3

[2] Tae Ha Park, Marcus Märtens, Gurvan Lecuyer, Dario Izzo, and Simone D'Amico. SPEED+: Next-generation dataset for spacecraft pose estimation across domain gap. In *IEEE Aerospace Conference*, 2022. 2